Z10.2
(cont'd)

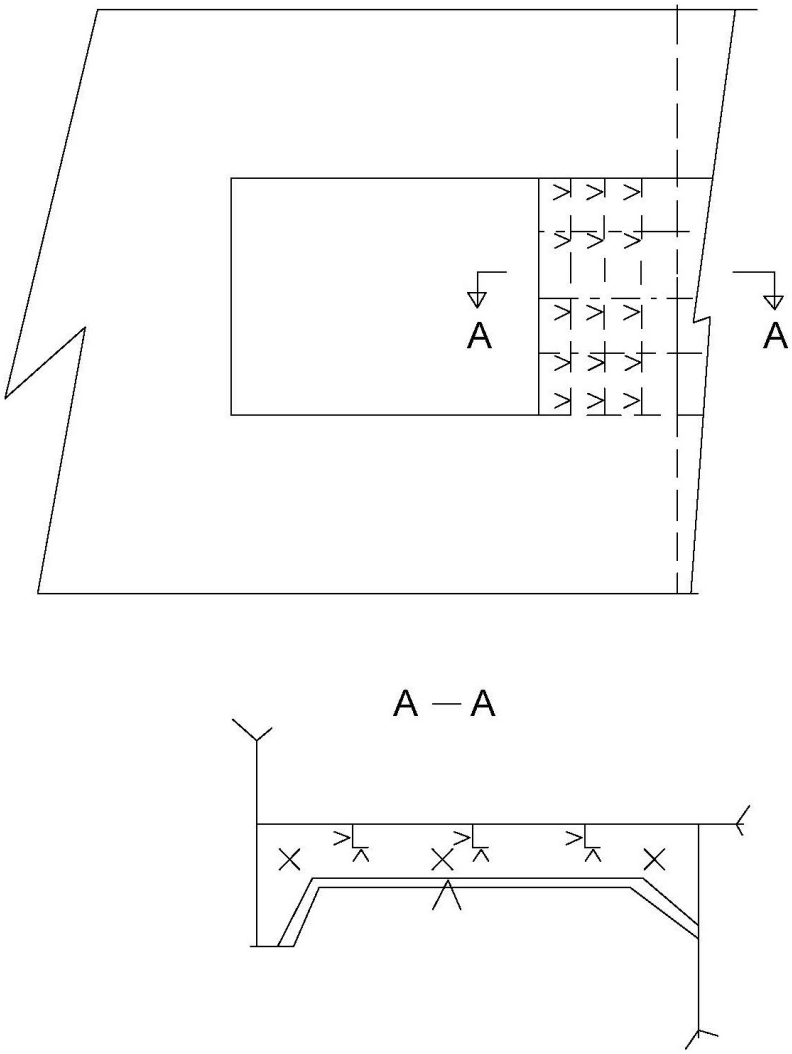

Figure 9 - Locations of measurements on underdeck structure

7.4 Reporting

7.4.1 A thickness measurement report is to be prepared. The report is to give the location of measurement, the thickness measured as well as corresponding original thickness. Furthermore, the report is to give the date when the measurement was carried out, type of measuring equipment, names of personnel and their qualifications and has to be signed by the operator.

The thickness measurement report is to follow the principles as specified in the Recommended Procedures for Thickness Measurement of Bulk Carriers, contained in Annex II.

7.4.2 The Surveyor is to review the final thickness measurement report and countersign the cover page.

Z10.2
(cont'd)

8. ACCEPTANCE CRITERIA

8.1 General

8.1.1 For vessels built under IACS Common Structural Rules, the Acceptance Criteria is according to IACS Common Structural Rules[1] and as specified in 8.2, 8.3 and 8.4.

8.1.2 For vessels not built under IACS Common Structural Rules, the Acceptance Criteria are according to the Rules of the individual Classification Society and/or specific IACS URs depending on ship's age and structural elements concerned, e.g UR S18 for corrugated transverse watertight bulkheads, UR S19 for the transverse watertight corrugated bulkhead between Cargo Holds Nos. 1 and 2, and URS21 for all cargo hatch covers and hatch forward and side coamings on exposed decks in position 1 (as defined in ILLC), UR S31 for side shell frames, as applicable.

8.2 Acceptance criteria for pitting corrosion of CSR ships

8.2.1 Side structures

If pitting intensity in an area where coating is required, according to IACS Common Structural Rules[2] is higher than 15% (see Figure 1), thickness measurements are to be performed to check the extent of pitting corrosion. The 15% is based on pitting or grooving on only one side of a plate.

In cases where pitting is exceeding 15%, as defined above, an area of 300 mm or more, at the most pitted part of the plate, is to be cleaned to bare metal and the thickness is to be measured in way of the five deepest pits within the cleaned area. The least thickness measured in way of any of these pits is to be taken as the thickness to be recorded.
The minimum remaining thickness in pits, grooves or other local areas is to be greater than the following without being greater than the renewal thickness (t_{ren}):

- 75% of the as-built thickness, in the frame and end brackets webs and flanges
- 70% of the as-built thickness, in the side shell, hopper tank and topside tank plating attached to the each side frame, over a width up to 30 mm from each side of it.

8.2.2 Other structures

For plates with pitting intensity less than 20%, see Figure 1, the measured thickness, t_m of any individual measurement is to meet the lesser of the following criteria:

$$t_m \geq 0.7\ (t_{as\text{-}built} - t_{vol\ add})\quad \text{mm}$$

$$t_m \geq t_{ren} - 1\quad \text{mm}$$

Where:

$t_{as\text{-}built}$ As-built thickness of the member, in mm

$t_{vol\ add}$ Voluntary thickness addition; thickness, in mm, voluntarily added as the Owner's extra margin for corrosion wastage in addition to t_C

[1] Ch.13 of IACS CSR for Bulk Carriers, or Ch. 13, Part 1 of IACS CSR BC&OT
[2] Sec. 5, Ch. 3 of IACS CSR for Bulk Carriers, or Sec. 4, Ch. 3, Part 1 of IACS CSR BC&OT

Z10.2
(cont'd)

t_{ren} Renewal thickness; minimum allowable thickness, in mm, below which renewal of structural members is to be carried out

t_C Total corrosion addition, in mm, defined in IACS Common Structural Rules[1]

t_m Measured thickness, in mm, on one item, i.e. average thickness on one item using the various measurements taken on this same item during periodical ship's in service surveys

The average thickness across any cross section in the plating is not to be less than the renewal criteria for general corrosion given in IACS Common Structural Rules[2].

8.3 Acceptance criteria for edge corrosion of CSR ships

8.3.1 Provided that the overall corroded height of the edge corrosion of the flange, or web in the case of flat bar stiffeners, is less than 25%, see Figure 2, of the stiffener flange breadth or web height, as applicable, the measured thickness, t_m, is to meet the lesser of the following criteria:

$$t_m \geq 0.7\ (t_{as\text{-}built} - t_{vol\ add})\ \text{mm}$$

$$t_m \geq t_{ren} - 1\ \text{mm}$$

8.3.2 The average measured thickness across the breadth or height of the stiffener is not to be less than that defined in IACS Common Structural Rules[2].

8.3.3 Plate edges at openings for manholes, lightening holes etc. may be below the minimum thickness given in IACS Common Structural Rules[2] provided that:

(a) the maximum extent of the reduced plate thickness, below the minimum given in IACS Common Structural Rules[2], from the opening edge is not more than 20% of the smallest dimension of the opening and does not exceed 100mm.

(b) rough or uneven edges may be cropped-back provided that the maximum dimension of the opening is not increased by more than 10% and the remaining thickness of the new edge is not less than t_{ren} - 1 mm.

8.4 Acceptance criteria for grooving corrosion of CSR ships

8.4.1 Where the groove breadth is a maximum of 15% of the web height, but not more than 30mm, see Figure 3, the measured thickness, t_m, in the grooved area is to meet the lesser of the following criteria:

$$t_m \geq 0.75\ (t_{as\text{-}built} - t_{vol\ add})\ \text{mm}$$

$$t_m \geq t_{ren} - 0.5\ \text{mm}$$

but is not to be less than

$$t_m = 6\ \text{mm}$$

[1] Sec. 3, Ch. 3 of IACS CSR for Bulk Carriers, or Sec. 3, Ch. 3, Part 1of IACS CSR BC & OT
[2] Ch. 13 of IACS CSR for Bulk Carriers, or Ch. 13, Part 1 of IACS CSR BC & OT

Z10.2
(cont'd)

8.4.2 Structural members with areas of grooving greater than those in 8.4.1 above are to be assessed based on the criteria for general corrosion as defined in IACS Common Structural Rules[1] using the average measured thickness across the plating/stiffener.

[1] Ch. 13 of IACS CSR for Bulk Carriers, or Ch. 13, Part 1 of IACS CSR BC & OT

Z10.2
(cont'd)

9. REPORTING AND EVALUATION OF SURVEY

9.1 Evaluation of Survey Report

9.1.1 The data and information on the structural condition of the vessel collected during the survey is to be evaluated for acceptability and continued structural integrity of the vessel.

9.1.1.1 For CSR bulk carriers, the ship's longitudinal strength is to be evaluated by using the thickness of structural members measured, renewed and reinforced, as appropriate, during the special surveys carried out after the ship reached 15 years of age (or during the special survey no. 3, if this is carried out before the ship reaches 15 years) in accordance with the criteria for longitudinal strength of the ship's hull girder for CSR bulk carriers specified in IACS Common Structural Rules[1].

9.1.1.2 The final result of evaluation of the ship's longitudinal strength required in 9.1.1.1, after renewal or reinforcement work of structural members, if carried out as a result of initial evaluation, is to be reported as a part of the Executive Hull Summary.

9.2 Reporting

9.2.1 Principles for survey reporting are shown in table VI.

9.2.2 When a survey is split between different survey stations, a report is to be made for each portion of the survey. A list of items examined and / or tested (pressure testing, thickness measurements etc.) and an indication of whether the item has been credited, are to be made available to the next attending surveyor(s), prior to continuing or completing the survey.

9.2.3 An Executive Hull Summary of the survey and results is to be issued to the Owner as shown in table VII and placed on board the vessel for reference at future surveys. The Executive Hull Summary is to be endorsed by the Classification Society's head office or regional managerial office.

[1] Ch. 13 of IACS CSR for Bulk Carriers, or Ch. 13, Part 1 of IACS CSR BC & OT

Z10.2
(cont'd)

TABLE I

TABLE OF MINIMUM REQUIREMENT FOR CLOSE-UP SURVEY AT SPECIAL HULL SURVEYS OF BULK CARRIERS

Special Survey No. 1 Age ≤ 5	Special Survey No. 2 5 < Age ≤ 10	Special Survey No. 3 10 < Age ≤ 15	Special Survey No. 4 and Subsequent Age > 15
(A) - 25% of shell frames in the forward cargo hold at representative positions. (A) - Selected frames in remaining cargo holds. (B) - One transverse web with associated plating and longitudinals in two representative water ballast tanks of each type (i.e. topside, or hopper side tank). (C) - Two selected cargo hold transverse bulkheads, including internal structure of upper and lower stools, where fitted. (D) - All cargo hold hatch covers and coamings (plating and stiffeners).	(A) - All shell frames in the forward cargo hold and 25% of shell frames in each of the remaining cargo holds, including upper and lower end attachments and adjacent shell plating. For Bulk Carriers 100,000 DWT and above, all shell frames in the forward cargo hold and 50% of shell frames in each of the remaining cargo holds, including upper and lower end attachments and adjacent shell plating. (B) - One transverse web with associated plating and longitudinals in each water ballast tank. (B) - Forward and aft transverse bulkhead in one ballast tank, including stiffening system. (C) - All cargo hold transverse bulkheads, including internal structure of upper and lower stools, where fitted. (D) - All cargo hold hatch covers and coamings (plating and stiffeners). (E) - All deck plating and under deck structure inside line of hatch openings between all cargo hold hatches.	(A) - All shell frames in the forward and one other selected cargo hold and 50% of frames in each of the remaining cargo holds, including upper and lower end attachments and adjacent shell plating. (B) - All transverse webs with associated plating and longitudinals in each water ballast tank. (B) - All transverse bulkheads in ballast tanks, including stiffening system. Areas (C), (D) and (E) as for Special Survey No. 2	(A) - All shell frames in all cargo holds including upper and lower end attachments and adjacent shell plating. Areas (B) - (E) as for Special Survey No. 3

(A) - Cargo hold transverse frames
(B) - Transverse web frame or watertight transverse bulkhead in water ballast tanks
(C) - Cargo hold transverse bulkheads plating, stiffeners and girders
(D) - Cargo hold hatch covers and coamings. Subject to cargo hold hatch covers of approved design which structurally have no access to the internals, close-up survey/thickness measurement shall be done of accessible parts of hatch covers structures.
(E) - Deck plating and under deck structure inside line of hatch openings between cargo hold hatches
See sketches of sheet 14 for the areas corresponding to (A), (B), (C), (D) and (E)
See also sketch in Annex V for zones of side shell frames for ships subject to compliance with UR S31
Note: Close-up Survey of transverse bulkheads to be carried out at four levels:
Level (a) Immediately above the inner bottom and immediately above the line of gussets (if fitted) and shedders for ships without lower stool.
Level (b) Immediately above and below the lower stool shelf plate (for those ships fitted with lower stools), and immediately above the line of the shedder plates.
Level (c) About mid-height of the bulkhead.
Level (d) Immediately below the upper deck plating and immediately adjacent to the upper wing tank, and immediately below the upper stool shelf plate for those ships fitted with upper stools, or immediately below the topside tanks.

Z10.2
(cont'd)

TABLE II

MINIMUM REQUIREMENTS FOR THICKNESS MEASUREMENTS AT SPECIAL HULL SURVEY OF BULK CARRIERS

Special Survey No. 1 Age ≤ 5	Special Survey No. 2 5 < Age ≤ 10	Special Survey No. 3 10 < Age ≤ 15	Special Survey No. 4 and Subsequent Age > 15
Suspect areas	Suspect areas	Suspect areas	Suspect areas
	Within the cargo length: Two transverse sections of deck plating outside line of cargo hatch openings	Within the cargo length: - each deck plate outside line of cargo hatch openings - two transverse sections, one in the amidship area, outside line of cargo hatch openings - all wind and water strakes	Within the cargo length: - each deck plate outside line of cargo hatch openings - three transverse sections, one in the amidship area, outside line of cargo hatch openings - each bottom plate
	Wind and water strakes in way of the two transverse sections considered above Selected wind and water strakes outside the cargo length area	Selected wind and water strakes outside the cargo length area	All wind and water strakes, full length
	Measurements, for general assessment and recording of corrosion pattern, of those structural members subject to close-up survey according to Table I	Measurements, for general assessment and recording of corrosion pattern, of those structural members subject to close-up survey according to Table I	Measurements, for general assessment and recording of corrosion pattern, of those structural members subject to close-up survey according to Table I
		See 1.1.4 and Annex III for additional thickness measurement guidelines applicable to the vertically corrugated transverse watertight bulkhead between cargo hold Nos. 1 and 2 on ships subject to compliance with URs S19 and S23.	See 1.1.4 and Annex III for additional thickness measurement guidelines applicable to the vertically corrugated transverse watertight bulkhead between cargo hold Nos. 1 and 2 on ships subject to compliance with URs S19 and S23.
	See 1.1.5 and Annex V for additional thickness measurement guidelines applicable to the side shell frames and brackets on ships subject to compliance with UR S31.	See 1.1.5 and Annex V for additional thickness measurement guidelines applicable to the side shell frames and brackets on ships subject to compliance with UR S31.	See 1.1.5 and Annex V for additional thickness measurement guidelines applicable to the side shell frames and brackets on ships subject to compliance with UR S31.

Z10.2
(cont'd)

TABLE III

Ship Name: .....

OWNERS INSPECTION REPORT - Structural Condition
For Tank/Hold No: .....

Grade of Steel:	Deck	:	Side	: .....
	Bottom	:	Long. Bhd	: .....

Elements Other	Cracks	Buckles	Corrosion	Coating cond.	Pitting	Mod. /Rep.

Deck:

Bottom:

Side:

Long. Bulkheads:

Transv. Bulkheads:

Repairs carried out due to:

Thickness measurements carried out, dates:
Results in General:

Overdue Surveys:

Outstanding Conditions of class:

Comments:

*Repairs are to be surveyed by the classification society

Date of Inspection: _____
Inspected by: _____
Signature: _____

Z10.2
(cont'd)

TABLE IV

PRINCIPLES FOR PLANNING DOCUMENT

Note: Table IV is superseded by Annex I: Guidelines for Technical Assessment in conjunction with planning for Enhanced Surveys of Bulk Carriers Special Survey - Hull.

Z10.2
(cont'd)

TABLE V

PROCEDURES FOR CERTIFICATION OF FIRMS ENGAGED IN THICKNESS MEASUREMENT OF HULL STRUCTURES

1. Application

This guidance applies for certification of the firms which intend to engage in the thickness measurement of hull structures of the vessels.

2. Procedures for Certification

(1) Submission of Documents:

Following documents are to be submitted to the society for approval:

a) Outline of firms, e.g. organisation and management structure.
b) Experiences of the firms on thickness measurement inter alia of hull structures of the vessels.
c) Technicians careers, i.e. experiences of technicians as thickness measurement operators, technical knowledge of hull structure etc. Operators, are to be qualified according to a recognized industrial NDT Standard.
d) Equipment used for thickness measurement such as ultra-sonic testing machines and its maintenance/calibration procedures.
e) A guide for thickness measurement operators.
f) Training programmes of technicians for thickness measurement.
g) Measurement record format in accordance with the Recommended Procedures for Thickness Measurements of Bulk Carriers, contained in Annex II.

(2) Auditing of the firms:

Upon reviewing the documents submitted with satisfactory results, the firm is audited in order to ascertain that the firm is duly organized and managed in accordance with the documents submitted, and eventually is capable of conducting thickness measurement of the hull construction of the ships.

(3) Certification is conditional on an onboard demonstration at thickness measurements as well as satisfactory reporting.

3. Certification

(1) Upon satisfactory results of both the audit of the firm in 2(2) and the demonstration tests in 2(3) above, the Society will issue a Certificate of Approval as well as a notice to the effect that the thickness measurement operation system of the firm has been certified by the Society.

(2) Renewal/endorsement of the Certificate is to be made at intervals not exceeding 3 years by verification that original conditions are maintained.

Z10.2
(cont'd)

4. Information of any alteration to the Certified Thickness Measurement Operation System

In case where any alteration to the certified thickness measurement operation system of the firm is made, such an alteration is to be immediately informed to the Society. Re-audit is made where deemed necessary by the Society.

5. Cancellation of Approval

Approval may be cancelled in the following cases:

(1) Where the measurements were improperly carried out or the results were improperly reported.

(2) Where the Society's surveyor found any deficiencies in the approved thickness measurement operation systems of the firm.

(3) Where the firm failed to inform of any alteration in 4 above to the Society.

Z10.2
(cont'd)

TABLE VI

SURVEY REPORTING PRINCIPLES

As a principle, for bulk carriers subject to ESP, the surveyor is to include the following content in his report for survey of hull structure and piping systems, as relevant for the survey.

The structure of the reporting content may be different, depending on the report system for the respective Societies.

1. General

1.1 A survey report is to be generated in the following cases:

- In connection with commencement, continuation and / or completion of periodical hull surveys, i.e. annual, intermediate and special surveys, as relevant

- When structural damages / defects have been found
- When repairs, renewals or modifications have been carried out
- When condition of class has been imposed or deleted

1.2 The purpose of reporting is to provide:

- Evidence that prescribed surveys have been carried out in accordance with applicable classification rules
- Documentation of surveys carried out with findings, repairs carried out and condition of class imposed or deleted
- Survey records, including actions taken, which shall form an auditable documentary trail. Survey reports are to be kept in the survey report file required to be on board
- Information for planning of future surveys
- Information which may be used as input for maintenance of classification rules and instructions

1.3 When a survey is split between different survey stations, a report is to be made for each portion of the survey. A list of items surveyed, relevant findings and an indication of whether the item has been credited, are to be made available to the next attending surveyor, prior to continuing or completing the survey. Thickness measurement and tank testing carried out is also to be listed for the next surveyor.

2. Extent of the survey

2.1 Identification of compartments where an overall survey has been carried out.

2.2 Identification of locations, in each ballast tank and cargo hold including hatch covers and coamings, where a close-up survey has been carried out, together with information of the means of access used.

2.3 Identification of locations, in each ballast tank and cargo hold including hatch covers and coamings, where thickness measurement has been carried out.

Note: As a minimum, the identification of location of close-up survey and thickness measurement is to include a confirmation with description of individual structural members corresponding to the extent of requirements stipulated in UR Z10.2 based on type of periodical survey and the ship's age.

Z10.2 (cont'd)

Where only partial survey is required, i.e. 25% of shell frames, one transverse web, two selected cargo hold transverse bulkheads, the identification is to include location within each ballast tank and cargo hold by reference to frame numbers.

2.4 For areas in ballast tanks and cargo holds where protective coating is found to be in GOOD condition and the extent of close-up survey and / or thickness measurement has been specially considered, structures subject to special consideration are to be identified.

2.5 Identification of tanks subject to tank testing.

2.6 Identification of piping systems on deck and within cargo holds, ballast tanks, pipe tunnels, cofferdams and void spaces where:

- Examination including internal examination of piping with valves and fittings and thickness measurement, as relevant, has been carried out

- Operational test to working pressure has been carried out

3. Result of the survey

3.1 Type, extent and condition of protective coating in each tank, as relevant (rated GOOD, FAIR or POOR).

3.2 Structural condition of each compartment with information on the following, as relevant:

- Identification of findings, such as:

  - Corrosion with description of location, type and extent
  - Areas with substantial corrosion
  - Cracks / fractures with description of location and extent
  - Buckling with description of location and extent
  - Indents with description of location and extent

- Identification of compartments where no structural damages / defects are found

The report may be supplemented by sketches / photos.

3.3 Thickness measurement report is to be verified and signed by the surveyor controlling the measurements on board.

4. Actions taken with respect to findings

4.1 Whenever the attending surveyor is of the opinion that repairs are required, each item to be repaired is to be identified in the survey report. Whenever repairs are carried out, details of the repairs effected are to be reported by making specific reference to relevant items in the survey report.

4.2 Repairs carried out are to be reported with identification of:

- Compartment
- Structural member
- Repair method (i.e. renewal or modification) including:
  - steel grades and scantlings (if different from the original);
  - sketches/photos, as appropriate;
- Repair extent

Z10.2
(cont'd)

- NDT / Tests

4.3 For repairs not completed at the time of survey, condition of class is to be imposed with a specific time limit for the repairs. In order to provide correct and proper information to the surveyor attending for survey of the repairs, condition of class is to be sufficiently detailed with identification of each item to be repaired. For identification of extensive repairs, reference may be given to the survey report.

Z10.2
(cont'd)

TABLE VII (i)

IACS UNIFIED REQUIREMENTS FOR ENHANCED SURVEYS
EXECUTIVE HULL SUMMARY

Issued upon Completion of Special Survey

GENERAL PARTICULARS

SHIP'S NAME:

CLASS IDENTIFY NUMBER:

IMO IDENTIFY NUMBER:

PORT OF REGISTRY:

NATIONAL FLAG:

DEADWEIGHT (M. TONNES):

GROSS TONNAGE:
NATIONAL:
ITC (69):

DATE OF BUILD:

CLASSIFICATION NOTATION:

DATE OF MAJOR CONVERSION:

TYPE OF CONVERSION:

a) The survey reports and documents listed below have been reviewed by the undersigned and found to be satisfactory

b) A summary of the survey is attached herewith on sheet 2

c) The hull special survey has been completed in accordance with the Regulations on [date]

Executive Summary Report completed by:	Name Signature	Title
OFFICE	DATE	
Executive Summary Report verified by:	Name Signature	Title
OFFICE	DATE	

Attached reports and documents:
1)
2)
3)
4)
5)
6)

Z10.2
(cont'd)

TABLE VII (ii)

EXECUTIVE HULL SUMMARY

A)	General Particulars:	-	Ref.Table VII (i)
B)	Report Review:	-	Where and how survey was done
C)	Close-up Survey:	-	Extent (Which tanks)
D)	Thickness measurements:	-	Reference to Thickness Measurement report
		-	Summary of where measured
		-	Separate form indicating the tanks/areas with Substantial Corrosion, and corresponding
		-	Thickness diminution
		-	Corrosion pattern
E)	Tank Protection:	-	Separate form indicating:
		-	Location of coating
		-	Condition of coating (if applicable)
F)	Repairs:	-	Identification of tanks/areas
G)	Conditions of Class:		
H)	Memoranda:	-	Acceptable defects
		-	Any points of attention for future surveys, e.g. for Suspect Areas.
		-	Examination of ballast tanks at annual surveys due to coating breakdown
I)	Conclusion:	-	Statement on evaluation/verification of survey report

Z10.2
(cont'd)

TABLE VII (iii) A – non CSR vessels

EXTRACT OF THICKNESS MEASUREMENT

Reference is made to the thickness measurements report:

1) Position of substantially corroded Tanks/Areas or Areas with deep pitting	Thickness diminution[%]	2) Corrosion pattern	Remarks: e.g. Ref. attached sketches

Remarks

1) Substantial corrosion, i.e. 75 – 100% of acceptable margins wasted.

2) P = Pitting
C = Corrosion in General
Any bottom plating with a pitting intensity of 20% or more, with wastage in the substantial corrosion range or having an average depth of pitting of 1/3 or more of actual plate thickness is to be noted.

Z10.2
(cont'd)

TABLE VII (iii) B – CSR vessels

EXTRACT OF THICKNESS MEASUREMENTS

Reference is made to the thickness measurements report:

1) Position of substantially corroded Tanks/Areas or Areas with deep pitting	$t_m - t_{ren}$ (mm)	2) Corrosion pattern	Remarks: e.g. Ref. Attached sketches

Remarks

1) Substantial corrosion, an extent of corrosion such that the assessment of the corrosion pattern indicates a measured thickness between t_{ren} + 0.5mm and t_{ren}.

2) P = Pitting
C = Corrosion in General
Areas with deep pitting assessed according to 8.2 are to be recorded in this column.

Z10.2
(cont'd)

TABLE VII (iv)

SPACE PROTECTION

1) Tank/hold Nos.	2) Tank/hold protection	3) Coating condition	Remarks

Remarks:

1) All ballast tanks and cargo holds to be listed.

2) C = Coating NP = No Protection

3) Coating condition according to the following standard

GOOD condition with only minor spot rusting.

FAIR condition with local breakdown at edges of stiffeners and weld connections and/or light rusting over 20% or more of areas under consideration, but less than as defined for POOR condition.

POOR condition with general breakdown of coating over 20% or more of areas or hard scale at 10% or more of areas under consideration.

For ballast tanks, if coating condition less than **"GOOD"** is given, tanks are to be examined at annual surveys. This is to be noted in part H) of the Executive Hull Summary.

Z10.2
(cont'd)

TABLE VIII
Sheet 1

REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENT AT THOSE AREAS OF SUBSTANTIAL CORROSION
SPECIAL SURVEY OF BULK CARRIERS WITHIN THE CARGO AREA

SHELL STRUCTURES

STRUCTURAL MEMBER	EXTENT OF MEASUREMENT	PATTERN OF MEASUREMENT
1. Bottom and Side Shell plating	a. Suspect plate, plus four adjacent plates b. See other tables for particulars on gauging in way of tanks and cargo holds	a. 5 point pattern for each panel between longitudinals
2. Bottom/Side Shell longitudinals	Minimum of three longitudinals in way of suspect areas	3 measurements in line across web 3 measurements on flange